Fiscal Year:	FY 2023	**Task Last Updated:**	FY 12/17/2022
PI Name:	Matson, Douglas Ph.D.		
Project Title:	Electromagnetic Levitation Flight Support for Transient Observation of Nucleation Events		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	douglas.matson@tufts.edu	**Fax:**	FY (617) 627-3058
PI Organization Type:	UNIVERSITY	**Phone:**	(617) 627-3239
Organization Name:	Tufts University		
PI Address 1:	Mechanical Engineering		
PI Address 2:	025 Anderson Hall, 200 College Ave		
PI Web Page:	https://		
City:	Medford	**State:**	MA
Zip Code:	02155	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	12/01/2015	**End Date:**	02/29/2024
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	4	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Sansoucie, Michael	**Contact Phone:**	256.544.5269
Contact Email:	michael.p.sansoucie@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: New grant number 80NSSC19K0256 awarded per NSSC information with period of performance of 3/1/2019-2/29/2024; this record for NNX16AB59G appended with that information per discussions with M. Sansoucie & L. Strutzenberg/MSFC (Ed., 6/21/19) NOTE: Extended to 11/30/2019 per NSSC information (Ed., 12/28/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K0256 ; NNX16AB59G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	EDITOR's NOTE (June 2019): New grant number 80NSSC19K0256 awarded per NSSC information with period of performance of 3/1/2019-2/29/2024; this record for NNX16AB59G (POP of 12/1/2015-2/28/2019) appended with that information per discussions with M. Sansoucie & L. Strutzenberg/Marshall Space Flight Center (MSFC). Continuation of "Electromagnetic Levitation Flight Support for Transient Observation of Nucleation Events" ; grant NNX14AB74G. See that project for previous reports. Funding is for Dr. Matson's role as Team Member on the subprojects described below, associated with European Space Agency (ESA)-sponsored main projects as noted. Main Project: Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC); Science Coordinator is Dr. Thomas Volkmann, German Aerospace Center DLR, Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany Subproject: Transformation kinetics - ISS (International Space Station) Using results from thermophysical property evaluation and combining these with the MHD convection definition, we will set convection over as wide a range as is possible and conduct recalescence studies from the undercooled melt using the Materials Science Laboratory-Electromagnetic Levitator (MSL-EML) to compare to results from ground-based EML and ESL. The critical parameters of interest are evaluation of the incubation time between formation of metastable phases and subsequent transformation to the stable phase for Fe-based and titanium aluminide alloys. Since this transformation is very rapid, the high-speed video system must be used to image the processes associated with nucleation and growth during phase selection. Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany Subprojects: Management and Coordination. Coordination of the scientific program, Coordination of the ground based measurement program, Coordination of the microgravity experiment program, Contact with the national and international space agencies. Short Duration Microgravity Measurements - PF. Conduct parabolic flights (PF) on selected alloys. Long Duration Microgravity Experiments - ISS. Surface tension and viscosity as a function of temperature, Density as a function of temperature, Solidification growth velocity. Preparation of the Microgravity Experiments. Experimental protocols for processing on the ISS ; Verification of flight samples for the ISS. MODELLING A: Thermodynamics, Phase Selection. Thermodynamic phase diagram calculation, Modeling of the influence of convection on phase selection.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The value of this research is that the results have application to the design of industrial welding, spray forming, and casting operations for a commercially important class of soft magnetic materials. In addition, this research addresses fundamental issues relating to rapid solidification behavior, metastable phase selection, and analysis of the processes governing microstructural evolution. Experiments focus on the use of convection as a processing control parameter.
Task Progress:	• Graduated 1 Ph.D. and 1 M.S. and 1 B.S. senior honors thesis student. • Published 1 peer-reviewed article, 1 book chapter, 2 thesis submissions, 1 art presentation, and 8 invited talks (see details in later section), plus 7 other papers in various stages of submission for next year. • Recognized as a Fellow for ASM-International (formerly known as the American Society for Metals) materials science professional society. • Served as Chair of the International Space Station National Laboratory (ISSNL) User Advisory Committee supporting Center for the Advancement of Science in Space (CASIS) operations (Ops). • Served as Co-Chair of the National Academies Committee on Biological and Physical Sciences in Space to oversee the 2023 NASA Decadal Study. • Conducted flight testing during ISS expedition 66 using the European Space Agency (ESA) International Space Station - Electromagnetic Levitator (ISS-EML) facility processing FeCrNi (#84 & #86) and CoSi (#52). • Conducted Inconel 625 and 718 testing using the Leibniz Institute for Solid State and Materials Research in Dresden (IFW-Dresden) portable EML at the German Synchrotron (Deutsches Elektronen-Synchrotron / DESY) in support of Batch 4 processing. • Virtually attended the 38th American Society for Gravitational and Space Research (ASGSR-38) 2022 Annual Conference, Houston, November 9-12 as organizer and as lead for the ISS-NL User Advisory Committee Public Meeting. • Featured speaking engagements at the 34th Japan Society of Microgravity Application (JASMA-34) event for 2021 Best Paper Award, Space Studies Board Panelist November 2022, plenary speaker for 2022 Asian Microgravity Symposium (AMS-2022). • Conducted flight planning for Batch 4 lead on Au sample (#122), Inconel 625 (#121), and Inconel 718 (#120) and for FeCo30 (#99) in collaboration with the ESA space PERITECTICA (international acronym PARSEC) program.
Bibliography Type:	Description: (Last Updated: 05/05/2026)
Abstracts for Journals and Proceedings	Nawer J, Kolbe M, Stanford B, Matson DM. "The moon and the stars." 38th Annual Meeting of the American Society for Gravitational and Space Research, Houston, TX, November 9-12, 2022. Abstracts. 38th Annual Meeting of the American Society for Gravitational and Space Research, Houston, TX, November 9-12, 2022. , Nov-2022
Articles in Peer-reviewed Journals	Bracker GP, Schneider S, Matson DM, Hyers RW. "Dynamic nucleation in sub-critically undercooled melts during electromagnetic levitation." Materialia. 2022 December;26:101623. https://doi.org/10.1016/j.mtla.2022.101623 , Dec-2022
Awards	Matson DM. "2022 Award of Fellow for ASM-International." May-2022

Awards	Matson DM, Xiao X, Rodriguez J, Wunderlich RK. "Japan Society of Microgravity Application (JASMA) Outstanding Paper Award (2021-22)." Sep-2022
Awards	Xiao X, Brillo J, Lee J, Hyers RW, Matson DM. "Nature Microgravity Paper featured as Editor's Pick of 2021 (2022)." Jan-2022
Awards	Wunderlich RK, Mohr M, Dong Y, Hecht U, Matson DM, Hyers RW, Bracker G, Lee J., Schneider S, Xiao X, Fecht H-J. "German Materials Society (DGM) Köster Award." Jun-2022
Books/Book Chapters	Matson DM. "Chapter 14 – Influence of convection on phase selection." in "Metallurgy in space: Recent results from ISS." Ed. H-J. Fecht, M. Mohr. Cham, Switzerland: Springer Cham, 2022. p. 299-314. https://doi.org/10.1007/978-3-030-89784-0 , Apr-2022
Dissertations and Theses	Cai Z. "Droplet deformation and oscillation in an acoustic levitator." MS Thesis, Mechanical Engineering Department, Tufts University, May 2022. , May-2022
Dissertations and Theses	Waskow D. "The effect of undercooling in an FeNi alloy." BS Honors Thesis, Mechanical Engineering Department, Tufts University, Feb 2023. , Feb-2023